



**Verified Carbon
Standard**

NKHOTAKOTA SOLAR PV PROJECT



south pole
The Climate Company

Project title	Nkhotakota Solar PV project
Project ID	4486
Crediting period	29-July-2023 to 28-July-2030
Original date of issue	18- July-2024
Most recent date of issue	06- February-2025
Version	4.0
VCS Standard Version	VCS Version 4.7
Prepared by	South Pole Carbon Asset Management Ltd

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1 PROJECT DETAILS

1.1 Summary Description of the Project

The proposed project activity is a greenfield, grid connected solar PV based electricity generation project, located in the Least Developed Country (UNCTAD¹) of Malawi, Africa. The purpose of this project is to provide clean energy to the local power grid as per design specifications and bolster renewable energy penetration in the relatively remote areas of Malawi, thereby improving energy access. This project is a voluntary project undertaken by the project developer Serengeti Nkhotakota Ltd. Further this project is not required or mandated the local laws of the Government of Malawi. It is configured as a single instance of project activity and is not a grouped project.

With an electrification rate of only around 11% (UNCTAD²), Malawi has one of the most severely constrained power sectors in sub-Saharan Africa. Total system capacity stands at just 532³ MW.

This Project entails the development of a 21 MWac / 26 MWp grid-connected solar PV plant, associated infrastructure including the sub-station and overhead power lines to connect to the Malawian ESCOM's 66 kV transmission line with the aim to bolster the Malawian grid's access to clean and renewable energy.

Pre project Scenario and baseline scenario are the same. The equivalent amount of electricity generated from the solar pv project would have been generated by the Malawian grid connected power plants encompassing fossil fuel-based power plants and stand by diesel power plants.

The project activity helps in reducing the GHG emission by replacing the equivalent quantity of fossil fuel fired grid connected power plants thereby, reduces 22,148 tCO₂e per year on average with a total of 155,036 tCO₂e over the first crediting period of 7 years.

The project site's coordinates are 12° 57'50.4"S 34° 15'19.9"E

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Validation/verification	(29-July-2023--28-July-2030)	VCS	EcoLance Pvt Ltd	7

1.3 Sectoral Scope and Project Type

Complete the table below with information relevant for non-AFOLU projects:

Sectoral scope ¹	01 Energy industries (renewable / non-renewables sources)
Project activity type	Type 1 : Renewable Energy Projects

1.4 Project Eligibility

1.4.1 General eligibility

As per the latest applicable United Nations Datasets⁵, Malawi is a Least Developed Country, therefore a persuasive case is made that the project is eligible under the scope of the VCS Program Standard (V 4.5) with reference to Section 2.1.1, 2.1.2 and 2.1.3 namely:

- It results in CO2 emission reductions, one of the seven Kyoto Protocol greenhouse gases.
- It consists in grid-connected electricity generation using a greenfield solar power plant in a Least Developed Country
- The project activity follows the approved methodology's definition of project area and boundary. The current project is a single installation of greenfield solar PV of 21 MWac capacity.
- The project is a greenfield solar PV project in an LDC with very low energy access, therefore the project has been developed with the purpose of creating renewable energy for injection into the Malawian Grid. As Malawi is Least Developed Country as designated by the United Nations, the project does not belong to the projects excluded in Table 1 of VCS Standard.

Further, the project's eligibility and applicability under the ACM0002 Methodology is explained in detail under Section 3.2 of this Project Design Document.

1.4.2 AFOLU project eligibility

Not Applicable

1.4.3 Transfer project eligibility

Not Applicable

1.5 Project Design

Indicate if the project has been designed as:

- ☒ Single location or installation

¹ Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: <https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes>

☐ Multiple locations or project activity instances (but not a grouped project)

☐ Grouped project

1.5.1 Grouped project design

Not Applicable

1.6 Project Proponent

Organization name	Serengeti Nkhotakota Limited
Contact person	Wilfred Van Den Bos
Title	Development Director
Address	Realty House, Churchill Road, Limbe, Malawi
Telephone	-
Email	wilfredvandenbos@serengetienergy.com

Organization name	South Pole Carbon Asset Management Ltd
Contact person	Mallika Bose
Title	Regional Director- South Asia and Middle East
Address	Technoparkstrasse 1, 8005 Zurich,Switzerland
Telephone	+41-43 501 35 50
Email	registration@southpole.com , pdc@southpole.com

1.7 Other Entities Involved in the Project

Not Applicable

1.8 Ownership

Serengeti Nkhotakota Limited owns the project and will demonstrate ownership of the land and the project with the requisite documents issued to them. The ownership of project activity can be verified through following documents from the government-authorised bodies: Commissioning Certificate & Energy generation statement issued by Electricity Supply Corporation of Malawi

(State Electricity Board). Commissioning Certificate & Energy generation statement issued by Electricity Supply Corporation of Malawi (State Electricity Board).

1.9 Project Start Date

Project start date	29-July-2023
Justification	Commissioning Certificate issued by Malawian Electricity Supply Commission (ESCOM)

1.10 Project Crediting Period

Crediting period	☒ Seven years, twice renewable ☐ Ten years, fixed ☐ Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)
Start and end date of first or fixed crediting period	29-July-2023 to 28-July-2031

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

Indicate the estimated annual GHG emission reductions/removals (ERRs) of the project:

- ☒ < 300,000 tCO₂e/year (project)
☐ ≥ 300,000 tCO₂e/year (large project)

Complete the table below for the first (if renewable) or fixed crediting period:

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO ₂ e)
29 th July 2023 to 31 st December 2023	9,227
1 st January 2024 to 31 st December 2024	22,148
1 st January 2025 to 31 st December 2025	22,148
1 st January 2026 to 31 st December 2026	22,148

1 st January 2027 to 31 st December 2027	22,148
1 st January 2028 to 31 st December 2028	22,148
1 st January 2029 to 31 st December 2029	22,148
1 st January 2030 to 31 st December 2030	22,148
Total estimated ERRs during the first or fixed crediting period	155,036
Total number of years	7
Average annual ERRs	22,148

1.12 Description of the Project Activity

The project uses photovoltaic solar power technology (PV). Under the current project, the set up involves 58,458 Suntech Hi Power model STP 445 solar PV panels to generate electricity by converting solar power into electric power to create clean energy with no consumption of fossil fuels and no emission of greenhouse gases. This is then converted from DC to AC using 132 string inverters of Huawei SUN2000- 105KTL-H1 make. The solar PV panels in each array are connected in parallel and/or in string to get the required current and voltage which match with the rated input of the inverter. The inverter connected in each array converts the DC current to AC current. The electricity collected from all the inverters is stepped up to 66 kV by using Step Up Transformer. It then is exported to the national grid through a metering system.

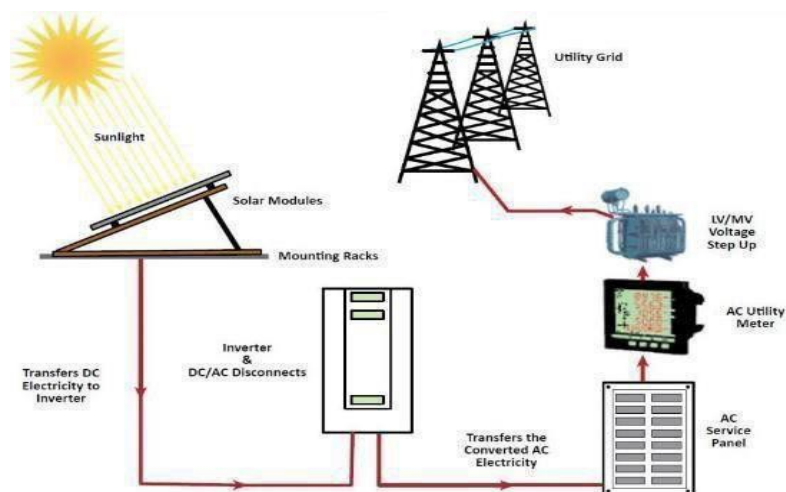


Image 1: Simple Single Line Diagram for project's source: detailed project report

1.13 Project Location

The site is located approximately 4km via road to the northwest of Nkhosakota, Malawi just off the M18 road between Nkhosakota and Kasungu. The site is approximately 60 hectares with overall dimensions of ~0.8km long by ~0.74km wide. With reference to below mentioned Image 2 and 3 the PV field is built within the rectangular section of the site.

The project site's coordinates are 12° 57' 50.4"S 34° 15' 19.9"E.



Image 3: Current project set up, source: project proponent

1.14 Conditions Prior to Project Initiation

The project is a green field solar PV installation, located in the Least Developed Country of Malawi (UNCTAD6). As per the US AID, the existing operating capacity of ESCOM of 532 MW, it is heavily dependent on hydro power, this is an issue as the country's dams frequently face drought and low water levels (USAID7). While most of the Malawi's power is hydro-based, efforts are underway to diversify generation sources with sources such as renewable solar PV and provide the local communities with energy.

A 'without project alternative' would place greater reliance on generating additional power from fossil fuel sources but also deny the local population potential employment opportunities associated with the Project. In this respect, the area would therefore continue along a marginalization path, with high poverty and unemployment levels, poor health facilities, poor access roads and poor state infrastructure.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project is not mandated by any law and regulations in Malawi, and this is a voluntary activity taken up by the project proponent.

Further, technology wise, renewable energy projects (including solar project) are not mandatory by the Government of Malawi².

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

If yes, provide required evidence of no double issuance as outlined by the VCS Standard.

1.16.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

If yes, provide the registration number and the date of project inactivity under the other GHG program.

Is the project active under the other program?

² <https://lexafrika.com/2024/05/renewable-enrgy-laws-malawi/>

☐ Yes ☒ No

Project proponents, or their authorized representative, must attest that the project is no longer active in the other GHG program in the Registration Representation.

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

If yes, provide the program name(s), reason(s) and date for the rejection, justification of eligibility under the VCS Program, and any other relevant information.

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

If yes, provide all required evidence of no double claiming as outlined by the VCS Standard.

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

If yes, provide all required evidence of no double claiming as outlined by the VCS Standard.

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

If yes:

Is the project proponent(s) or authorized representative a buyer or seller of the product(s) (goods or services) that are part of a supply chain?

☐ Yes ☐ No

If yes:

Has the project proponent(s) or authorized representative posted a public statement on their website saying, "Carbon credits may be issued through Verified Carbon Standard project [project ID] for the greenhouse gas emission reductions or removals associated with [project proponent or authorized representative organization name(s)] [name of product(s) whose emissions footprint is changed by the project activities]."

☐ Yes

☐ No

If yes to all:

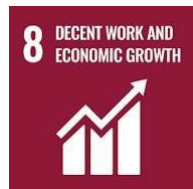
Provide evidence of the public statement. Evidence must be provided in this section or in an appendix.

1.18 Sustainable Development Contributions

The project works on renewable energy in Malawi which is a Least Developed Country. Further the specifically selected site has very low levels of electrification, industry and formal employment. The Nkhotakota project will provide clean and renewable energy to the local populace and the national grid serviced by it, thereby reducing the use of diesel Peaker power plants while expanding the role of renewables. This will lead to better health, environmental and industrial outcomes as detailed below:



The project will install a greenfield Solar PV power plant in a remote region of Malawi and bolster the National Grid of Malawi (ESCOM8) with clean and renewable energy. The project will achieve a GHG reduction of 22,148 tCO₂e/year during the first crediting period and therefore contribute to SDG 13 Climate Action



The construction and operation phase of this project will create new high-quality jobs for which people will be drawn from the nearby communities, which will have a positive impact for the local economy. Therefore, the project will achieve SDG 8 Decent Work and Economic Growth. Currently 25 new jobs have been created for the local communities.



As per the current project design and layout the project will generate an average of 56 GWhrs annually⁹ of clean, renewable energy over the first crediting period, providing the local communities with affordable energy access thereby achieving SDG 7 Affordable and Clean Energy.

1.19 Additional Information Relevant to the Project

1.19.1 Leakage Management

As this is a solar PV project there is no leakage anticipated.

1.19.2 Commercially Sensitive Information

No commercially sensitive information is part of this PDD.

1.19.3 Further Information

No further information is anticipated.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification	The stakeholders were identified using public notices, advertisements in English and Local vernacular print media and by notices pasted in prominent public places of the villages nearest the project site. Further, written invitations were issued to government officials and community leaders of those directly affected by the project /directly affected by the project and those indirectly affected by project. Further, in line with VERRA guidelines, the stakeholders consulted include: • Department of Environmental Affairs (DEA) • Malawian Investment Trade Centre (MITC) • Nkhosakota District Commissioner's (DC) Office • Representative from Nkhosakota District Lands Office • Local Traditional Authority (TA) • Senior Group Village Head (SGVH) (Head of the Village Heads) • Members of Local Villages
Legal or customary tenure/access rights	There are no legal or customary tenure access rights for local stakeholders identified.
Stakeholder diversity and changes over time	No changes are expected, as the stakeholders identified for the project activity are not diverse in nature.
Expected changes in well-being	It is expected that with the solar PV project, employment opportunities for local stakeholders will increase as the project employees members of the local community, further with the clean energy generated and supplied to the local grid, it is expected that energy access will increase for local stakeholders and community members.
Location of stakeholders	The nearest community members reside within a few hundred meters of the solar PV project.
Location of resources	-

2.1.2 Stakeholder Consultation and Ongoing Communication

Date of stakeholder consultation	Initial LSC: 17-January-2018
Stakeholder engagement process	Initial consultations as a part of the procedure followed by the project proponent were carried out at a meeting held in Nkhosakota on 17th January 2018, between representatives of the local District Commissioner (DC), MITC representatives, the local Traditional Authority (TA), Senior

	Group Village Head (SGVH) and the local Village Heads. The meeting was also attended by the Project Proponent team. The meeting served to provide stakeholders with information about the Project and hear the local communities' perspectives for the project.
Consultation outcome	Public meetings–These involved consultation and engagement with residents from local community members, village chiefs and other interest groups. Focus group discussions–These involved consultation and engagement with village chiefs, a women's group, an HIV support group, a men's group, a youth group and a business forum. Field observations- The physically inspected major facilities and landmarks, such as water supply points, school buildings, health care facilities, markets, occupational activities, informal sector activities and conditions of roads and houses. Other areas of interest studied were cultural heritage features of the project affected/neighboring communities. The demographic pattern, land size/use, economic activities (with emphasis on low-income groups highly dependent on primary activities), community structure, employment, markets, labor supply, income distribution, consumption, and migration pattern of the host/affected and neighboring communities was also discussed.
Ongoing communication	1) Grievance box installed at project site and communities, this acts as ongoing communication 2) 2nd LSC conducted 3) Livelihood Restoration Opportunities agreed by community These 2 demonstrate the continuous communication mechanism
Stakeholder input	-

Date of stakeholder consultation	Second LSC : 12-October-2023
Stakeholder engagement process	All the formal invitations for Government Officials were issued in writing by 6 th October 2023 in advance of the actual date of the LSC to ensure maximum awareness. Public invitations were issued on 26th September 2023 via newspaper advertisements in both English and Chechewa language publications to ensure widest participation of stakeholders affected by the project activity. The invitations were further posted at prominent public spaces near the project site including but not limited village schools, public squares and village health centers to ensure widest coverage of the local community's participation.

Consultation outcome	The LSC was conducted on 12th October with the project team explaining the Nkhotakota solar project's different aspects. Subsequently, the stakeholders aired their questions and concerns which were meticulously recorded. The detailed written queries of the stakeholders and their responses by the project team have been recorded and are submitted with this PD as Annexure -1
Ongoing communication	Grievance registers have been installed at project site as demonstrated during OSV along with grievance registers installed at the nearby communities.
Stakeholder input	Detailed Minutes of the Meet along with attendance sheets were taken during the LSC and the same is attached with this PD.

2.1.3 Free Prior and Informed Consent

Obtaining consent	The project proponent has leased the land from the landowners with rent receipt attached as part of this PDD.
Outcome of FPIC	FPIC has been taken and landowners paid rent.

2.1.4 Grievance Redress Procedure

Development process	In line with the outcomes of the LSC, grievance boxes have been installed at public spaces both at the project site and the nearby communities from where the stakeholders were identified
Grievance redress procedure	Grievance boxes have been installed at public spaces where aggrieved stakeholders can anonymously deposit their grievance. This box is checked every week and any issue identified is investigated by the management of the project and steps taken to rectify the situation.

2.1.5 Public Comments

Under ESIA's comments, a grievance mechanism was agreed implemented under the Environmental and Social Management Plan¹¹. This will ensure that all local stakeholders facing any issues can register their issue for resolution at any point of time throughout the lifetime of the project. The PP has installed grievance registers at key project sites across the Nkhotakota Solar Power project and near by community spaces.

No comments were received during the Project's Public Comment Period from its listing on the VERRA Registry page from 17th August 2023 to 16th September 2023.

Comments received	Actions taken
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Commissioning dates need to be improved	Commissioning dates fixed by ESCOM based on SoPs and same communicated to community.
Economic well being and community benefits can be elaborated	Local wage employment opportunities increased for community members through jobs.

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

Our project management team possesses expertise to run and implement this kind of project activities as their technical team possesses extensive experience in running similar renewable energy projects in African region2.

2.2.2 Risk Assessment

	Risks identified	Mitigation or preventative measure(s) taken
Natural and human-induced risks to stakeholders' wellbeing	No risks identified as this is a solar PV project	NA as no risk is identified.
Risks to stakeholder participation	No risks identified as this is a solar PV project	NA as no risk is identified.
Working conditions	HSE safeguarding and officer are present on site,	PPE kit is worn, HSE Officer present, as demonstrated during the validation site visit.
Safety of women and girls	A substantial % of female employees are on site. There is a continuous security coverage, this can further ensure there are no safety related issues for any gender	Continuous security coverage in the premises
Safety of minority and marginalized groups, including children	There is no child labor on project, hence not applicable	NA as there is no child labor at the project site.
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous	End of Life solar PV panels (after 25 years)	PP has already received a General Waste Management License from GoM

materials and
chemical pesticides
and fertilizers)

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

	Risks identified ³	Mitigation or preventative measure(s) taken
Discrimination	Written policies laid down in line with local regulations. Hence, no risk is identified.	Written policies laid down in line with local regulations
Sexual harassment	As per local regulations, sexual harassment is prohibited. Hence, no risk is identified.	Written policies laid down in line with local regulations
Equal pay for equal work	Gender balance ensured through employees present. Hence, no risk is identified.	Written policies laid down in line with local regulations
Gender equity in labor and work	In line with local regulations, pursuing all the compliances, gender equality maintained, a no. of female employees are present. Hence, no risk is identified.	Written policies laid down in line with local regulations
Forced labor	As per the local regulations, no forced labor are employed. Hence, no risk is identified.	Written policies laid down in line with local regulations
Child labor	As per the local regulations, child labor is prohibited. Hence, no risk is identified.	Written policies laid down in line with local regulations
Human trafficking	As per the local regulations, human trafficking is prohibited. Hence, no risk is identified.	Written policies laid down in line with local regulations

2.3.2 Human Rights

Risks identified	Mitigation or preventative measure(s) taken
No Risk identified	Written policies written

³ The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Mitigation(s) or preventative measure taken
Co benefits to local communities	Local community members employed during the construction and operation of this project

2.3.4 Property Rights

Risks identified	Mitigation or preventative measure(s) taken
Land ownership	Project land taken on rent with clearly defined Lease Agreement.

2.3.5 Benefit Sharing

Process used to design the benefit sharing plan	Serengeti is sole owner of this project; therefore, no benefit sharing is applicable
Summary of the benefit sharing plan	Not applicable
Approval and dissemination of benefit sharing plan	Not applicable

2.4 Ecosystem Health

	Risks identified	Mitigation or preventative measure(s) taken
Impacts on biodiversity and ecosystems	No risk	Solar PV project hence no risk identified
Soil degradation and soil erosion	No risk	Solar PV project hence no risk identified
Water consumption and stress	No risk	Solar PV project hence no risk identified

2.4.1 Rare, Threatened, and Endangered Species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☐ Yes ☒ No

If yes, list such species and habitats in the table below and provide evidence that the project will not adversely impact these areas.

Species and habitat	N/A
Areas needed for habitat connectivity	N/A

	Risks identified	Mitigation or preventative measure(s) taken
Habitats for rare, threatened, and endangered species	No Risk Identified	As project is a solar PV project, no risks to local species is identified
Areas for habitat connectivity	N/A	N/A

2.4.2 Introduction of Species

Species introduced	Classification	Justification for use	Adverse effects and mitigation
N/A	N/A	N/A	N/A

Existing invasive species	Mitigation measures to prevent the spread or continued existence of invasive species
N/A	N/A

	Risks identified	Mitigation or preventative measure(s) taken
Invasive species	N/A	N/A

2.4.3 Ecosystem Conversion

	Risks identified	Mitigation or preventative measure(s) taken
Ecosystem conversion	N/A	N/A

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Large Scale consolidated methodology	ACM0002	Grid-connected electricity generation from renewable sources	Version 21.0
Methodological tool7	Tool 07	Tool to calculate the emission factor for an electricity system.	Version 7.0

3.2 Applicability of Methodology

Methodology ID	Applicability condition	Justification of compliance
ACM0002	This methodology is applicable to grid connected renewable energy power generation project activities that: a) Install a Greenfield power plant; b) Involve a capacity addition to (an) existing plant(s); c) Involve a retrofit of (an) existing operating plants/units; d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or e) Involve a replacement of (an) existing plant(s)/unit(s)	The project is a greenfield grid connected renewable power generation project.
	In case the project activity involves the integration of a BESS, the methodology is applicable to grid-connected renewable energy power generation project activities that:	The project is a greenfield solar PV power project

	a) Integrate BESS with a Greenfield power plant; b) Integrate a BESS together with implementing a capacity addition to (an) existing solar photovoltaic or wind power plant(s)/unit(s); c) Integrate a BESS to (an) existing solar photovoltaic or wind power plant(s)/unit(s) without implementing any other changes to the existing plant(s); d) (d) Integrate a BESS together with implementing a retrofit of (an) existing solar photovoltaic or wind power plant(s)/unit(s).	
	The methodology is applicable under the following conditions: (a) Hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; (b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion, retrofit,	Not applicable, the project is a newly built solar power project.

or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity;

(c) In case of Greenfield project activities applicable under paragraph 5 (a) above, the project participants shall demonstrate that the BESS was an integral part of the design of the renewable energy project activity (e.g. by referring to feasibility studies or investment decision documents);

(d) The BESS should be charged with electricity generated from the associated renewable energy power plant(s). Only during exigencies may the BESS be charged with electricity from the grid or a fossil fuel electricity generator. In such cases, the corresponding GHG emissions shall be accounted for as project emissions following the requirements under section 5.4.4 below. The charging using the grid or using fossil fuel electricity generator should not amount to more than 2 per cent of the electricity generated by the project renewable energy plant during a monitoring period. During the time periods (e.g. week(s), months(s)) when the BESS consumes more than 2 per cent of the electricity for charging, the project participant

	shall not be entitled to issuance of the certified emission reductions for the concerned periods of the monitoring period.	
	In case of hydro power plants, one of the following conditions shall apply: (a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or (b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density, calculated using equation (7), is greater than 4 W/m²; or (c) The project activity results in new single or multiple reservoirs and the power density, calculated using equation (7), is greater than 4 W/m²; or (d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4 W/m², all of the following conditions shall apply: I. The power density calculated using the total installed capacity of the integrated project, as per equation	Not applicable, the project is a newly built solar power project.

	(8), is greater than 4 W/m² ; II. Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity; III. (iii) Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m² shall be: a. Lower than or equal to 15 MW; and b. Less than 10 per cent of the total installed capacity of integrated hydro power project.	
	In the case of integrated hydro power projects, project participants shall: (a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or (b) Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the	Not applicable, the project is a newly built solar power project.

	optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum of five years prior to the implementation of the CDM project activity.	
	The methodology is not applicable to: (a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; (b) Biomass fired power plants/units.	Not applicable, the project is a newly built solar power project.
	In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is “the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance”.	Not applicable, the project is a newly built solar power project.

Methodology ID	Applicability condition	Justification of compliance
Tool 07	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	This tool is applicable for this project activity, as the solar project supplies electricity to the grid.
	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability	The tool is applicable and the emission factor calculation is provided in the section 4.1 of the VCS PD.

	of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	
	In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	This criteria is not applicable, as the project activity is not located in Annex-I country.
	Under this tool, the value applied to the CO ₂ emission factor of biofuels is zero.	This is a solar PV project, hence, this criteria is not applicable for this project activity.

3.3 Project Boundary

According to ACM0002 (Version 21.0), the spatial extent of the project boundary includes the project power plant/unit and all power plants/units connected physically to the ESCOM (Malawian National Grid Operator) that the project power plant is connected to.

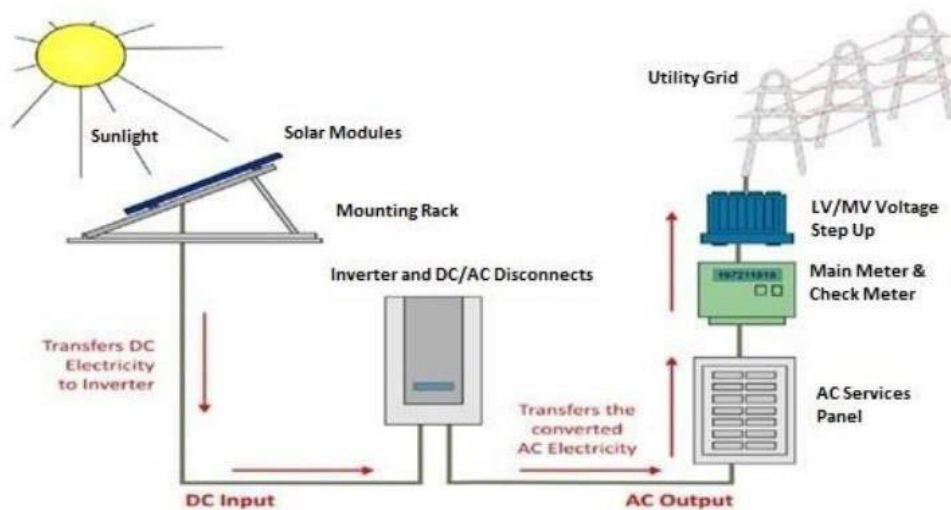


Image5 Simplified line diagram, source: project Detailed Project Report

Source	Gas	Included?	Justification/Explanation
Baseline	CO ₂	Yes	Main emission source from fossil fuel power plants
	CH ₄	Excluded	Excluded for simplification. This emission source is assumed to be very small.

Source		Gas	Included?	Justification/Explanation
		N ₂ O	Excluded	Excluded for simplification. This emission source is assumed to be very small.
		Other	Excluded	Excluded for simplification. This emission source is assumed to be very small.
	Source 2	CO ₂	Excluded	-
		CH ₄	Excluded	-
		N ₂ O	Excluded	-
		Other	Excluded	-
Project	Source 1	CO ₂	Excluded	No emissions from project
		CH ₄	Excluded	As per the ACM0002, the production and operation of solar photovoltaic projects do not produce significant greenhouse gas emissions.
		N ₂ O	Excluded	-
		Other	Excluded	-
	Source 2	CO ₂	Excluded	Thus, the project emissions can be safely ignored.
		CH ₄	Excluded	-
		N ₂ O	Excluded	-
		Other	Excluded	-

3.4 Baseline Scenario

According to the ACM0002 Large Scale Consolidated Methodology Version 21: “If the project activity is the installation of a Greenfield power plant. The baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as mentioned in the combined margin (CM) calculations described in “TOOL07: Tool to calculate the emission factor for an electricity system.” In the absence of the project activity, the equivalent amount of power would have been drawn from the Malawian grid or generated from fossil fuels. Hence, the baseline for the project activity is the equivalent amount of power from the national grid.

Project activity is the generation of electricity from renewable energy sources. The power generated from the solar PV plant has zero emissions; there is no material leakage and the electricity generated will be fed into the fossil intensive national grid through the interconnection facility at the site.

3.5 Additionality

Demonstrate and assess the additionality of the project, in accordance with the applied methodology and any relevant tools, taking into account the following additionality methods:

3.5.1 Regulatory Surplus

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes ☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☐ Yes ☒ No

If no, describe which mandated laws, statutes, or other regulatory frameworks require project activities and provide evidence of systematic non-enforcement to demonstrate regulatory surplus.

3.5.2 Additionality Methods

According to the methodology ACM0002 (Version 21.0), the additionality of the project activity shall be demonstrated and assessed using the latest version of the “TOOL01: Tool for the demonstration and assessment of additionality”. All calculations made are provided as annexures.

The steps are listed as following:

Step 0: First of Its Kind project activities, this does not apply as there are other solar projects within the project country.

Step 1: Identification of alternatives to the project activity consistent with current laws and regulations. Define realistic and credible alternatives to the project activity(s) through the following Sub-steps.

Sub-step 1a: Define alternatives to the project activity: Alternatives available to the project participant or similar project developers that provide outputs or services comparable with the project activity are identified below:

- (a) The proposed project activity undertaken without being registered as a VCS project activity.
Alternative
- (b) Continuation of the current situation (no project activity or other alternatives undertaken).

Sub-step 1b: Consistency with mandatory laws and regulations

Both alternatives (a) and (b) are realistic alternatives to the project, which are consistent with mandatory laws and regulations of Malawi.

Step 2: Investment analysis Sub-step 2a: Determine appropriate analysis method:

According to ACM0002 (Version 21.0), if the project activity and the alternatives identified in Step 1 generate no financial or economic benefits other than carbon related income, then apply the simple cost analysis (Option I). Otherwise, use the investment comparison analysis (Option II) or the benchmark analysis (Option III). For the project, the alternative to the project activity is the supply of electricity from a grid which is not considered as an investment by project participant, so the benchmark analysis (Option III) is used.

Below are the major chronology of events for the project activity:

Table 1 : Chronology of Events

Milestones	Date
Company incorporation	2 May 2018
Environmental Impact Assessment	29 June 2019
Loan Commitment Letter	20 December 2018
Third party due diligence report (DPR)	January 2019
Signing of Power Purchase Agreement	14 February 2019
EPC contract	14 April 2020
O & M Contract	9 June 2020
Commissioning of the project	29 July 2023

Sub-step 2b: Option III. Apply Benchmark Analysis

According to “Investment analysis” (version 13.0, in force during November 2023), “For Malawi, the rate 24.01% for Group 1 project (renewable power projects). 12

The latest version of the tool (Version 13.0) has been used.

As per paragraph 16 of Appendix A of the above-mentioned document, "In situations where an investment analysis is carried out in nominal terms, project owners can convert the real term values provided in the table below to nominal values by adding the inflation rate. The inflation rate shall be obtained from the inflation forecast of the central bank of the host country for the duration of the crediting period. If this information is not available, the target inflation rate of the central bank shall be used. If this information is also not available, then the average forecasted inflation rate for the host country published by the IMF (International Monetary Fund World Economic Outlook) or the World Bank for the next five years after the start of the project activity shall be used.

As per para 19 of EB 105, Annex 6 the cost of equity is determined by selecting the values provided in the Appendix, i.e., Default values for cost of equity (expected return on equity) is presented below:

The Required return on equity (benchmark) was computed in the following manner:

$$\text{Nominal Benchmark} = (1 + \text{Real Benchmark}) \cdot (1 + \text{Inflation rate}) - 1$$

Sub-step 2c: Calculation and comparison of financial indicators:

The relevant financial analysis is an Internal Rate of Return calculation from the project's costing data. The equity IRR is therefore calculated on the parameters that are applicable to the Malawian market data, this will be considering the specific characteristics of the project activity and not linked to the profitability for the project. Based on the parameters including the financial parameters, loan, working capital, applicable corporate taxes, the EIRR sheet was developed, the benchmark equity IRR for Malawi for the year 2023 was determined to be 31.02% and the project's was estimated to be 12.78%. This has been arrived at by considering the default values mentioned under Tool 27 Version 13.0 group 1 and adding the long term inflation forecast issued by the IMF for the specific years of 2023-2026.

Table 2 : IRR Sheet Parameters

Parameters	Values	Source
Location where the project is situated	Nkhotakota Malawi	
Total Capacity in AC (MW)	21.0	PPA
Total Capacity in DC (MWp)	26.000	PPA

Grid Injection is allowed for Capacity (AC)	21.000	PPA
Expected Date of Commissioning	29-Jul-23	As per commissioning certificate
Duration of PPA (Yrs.)	20	As per PPA page no 89
Generation and sale of electricity		
PLF (%)	24.79%	Calculated
Annual generation (kWh)@P50	5,64,58,000	Third party Report
Annual Degradation (%)	0.45%	Third party Report
Tariff (USD \$/kWh)	0.0972	PPA (P.No. 89)
Index on the PPA price per annum	0	PPA (P.No. 89)
Operation and maintenance cost and Insurance		
Opex (Mn. USD \$)	0.64	Year 1 Opex (Declaration by PP)
Index CPI on the OPEX (inflation rate)	2.00%	Declaration by PP
Financial parameters		
TOTAL COST (Mn. USD \$)	45.80	Audited statement
Equity Investment (Mn. USD \$)	11.45	Calculated
Loan Amount (Mn. USD \$)	28.40	Loan commitment letter
Term loan		
Equity (%)	25.00%	Loan commitment letter
Loan Amount (Mn. USD)	28.40	Loan commitment letter Annex A, page 2
Interest rate (%)	3.5%	Commitment Letter
Loan Tenure (Qtr.)	68	Loan commitment letter Annex A, page 2
Moratorium Period (Qtr.)	2	Loan commitment letter Annex A, page 2
Repayment Period (Qtr.)	66	Loan commitment letter Annex A, page 2
Repayment instalments value (Mn. USD)	0.430	Calculated
1st instalment from (Qtr. end)	31-Mar-24	
Book Depreciation (SLM Method)		
Land Cost (USD Mn.)	-	Third party Report
Gross Depreciable Value (Mn. USD)	45.80	Calculated
Depreciation (Mn. USD)	1.83	Calculated

Final Tax rates		
Corporate tax rate (%)	15.00%	https://taxsummaries.pwc.com/malawi/corporate/taxes-on-corporate-income

The project activity cannot be considered as financially attractive as the equity IRR for the project activity is less than the Malawian Benchmark EIRR.

Table 3 Estimated Project IRR and Benchmark Values :

Investment Analysis – Outcome	Equity IRR	Benchmark (Equity IRR)
	12.78 %	31.02%

Table 4 Sensitivity Analysis

Variation %	-10%	Normal	10%	Variation required to reach benchmark
PLF	9.87%	12.78%	16.88%	43.10%
Operations and Maintenance	13.46%	12.78%	12.10%	-271%
Project Costs	14.09%	12.78%	11.70%	-59.10%
Tariff	8.87%	12.78%	16.88%	99.50%

Sub-step 2d: Sensitivity Analysis The following parameters were used to develop the sensitivity analysis for this project:

1. PLF
2. O&M Cost
3. Project Cost
4. Tariff

Based on the above parameters, the ultimate objective of the sensitivity analysis is to determine the likelihood of the occurrence of a scenario other than the scenario presented, in order to provide a crosscheck on the suitability of the assumptions used in the development of the investment analysis, the results are below:

Table 5: Sensitivity Analysis

Variation %	-10%	Normal	10%	Variation required to reach benchmark
PLF	9.87%	12.78%	16.88%	43.10%
O and M	13.46%	12.78%	12.10%	-271%
Project Cost	14.09%	12.78%	11.70%	-59.10%

Tariff	8.87%	12.78%	16.88%	99.50%
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The results of sensitivity analysis show that even with a variation of +10% & -10% in project cost, O&M cost PLF and tariff rate, the Equity IRR is significantly lower than the benchmark. This is evident from the results given above; the project remains additional even under the most favorable conditions. This is elaborated below as:

- i) Tariff Rate: As the PPA is of fixed price for 25 years, there is no possibility of further variation (99.50%) of tariff which would result an IRR value which would cross the Benchmark.
- ii) Project Cost: As the project has been substantially delayed in it's commissioning, a downward variation (59.10%) in project costs is not possible.
- iii) Operation and Maintenance: As the Operation and Maintenance contract has been signed, any further downward variation (271%) is not expected.
- iv) PLF: As PLF value has been sourced from third party report, which has specified the maximum attainable generation for the entire life span of the project, therefore, a further variation (43.10%) of the PLF value to cross the benchmark is not a possible scenario.

Step 4: Common practice analysis As the project applies power generation based on renewable energy which is one of the measure(s) listed in the definitions section above, we proceed to Sub-step 4a.

Sub-step 4a: The proposed project activity(ies) applies measure(s) that are listed in the definitions section above Common practice analysis is carried out as per "Common practice" (version 03.1)

Sub-Step 4a-1: calculate applicable capacity or output range as +/-50% of the total design capacity or output of the proposed project activity The installed capacity of the project is 21 MW, so the applicable capacity range as +/-50% of the total design capacity.

Step 4a-2: identify similar projects which fulfil all of the following conditions: Malawi is selected as applicable geographical area. No projects are identified which started commercial operations before within the required timeframe.

We have taken solar PV as the technology measure as tariff structure and policy framework are different for different types of RE technologies.

Sub-step 4a-3: Within the projects identified in Sub-step 4a-2, identify those that are neither registered CDM/VCS/Gold Standard project activities, project activities submitted for registration, nor project activities undergoing validation. Note their number as Nall.

According to Sub-Step 4a-2, there are no solar power plants with capacity from 10.5 MW to 31 MW operational in Malawi, $N_{all} = 0$.

Sub-step 4-4: Within similar projects identified in Sub-step 4-3, identify those that apply technologies that are different to the technology applied in the proposed project activity. Note their number N_{diff} .

For the project, $N_{diff}=0$.

Sub-step 4-5: Calculate factor $F=1-N_{diff}/N_{all}$ representing the share of similar projects (penetration rate of the measure/technology) using a measure/technology like the measure/technology used in the proposed project activity that deliver the same output or capacity as the proposed project activity.

Since $N_{all} = 0$, $N_{diff}=0$, therefore, $N_{all} - N_{diff} = 0$, which is equal to 0. $F=1-N_{diff}/N_{all}=1$.

According to common practice (version 03.1), the proposed project activity is a “common practice” within a sector in the applicable geographical area if the factor F is greater than 0.2.

The $N_{all} - N_{diff}$ is 0, therefore less than 3. Thus, the Project activity is not a “common practice” as it satisfies the criteria laid.

Outcome of Step 4: Based on the analysis above, the baseline scenario of the project is: The proposed project can be considered as additional.

3.6 Methodology Deviations

There is no deviation from methodology

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

According to the latest version of the methodology, ACM0002 Large Scale Consolidated Methodology: Grid-connected Electricity Generation from Renewable Sources, Version 21.0, since the project activity is the installation of Greenfield Renewable Energy Power Plant, the baseline scenario is as follows: If the project activity is the installation of a Greenfield power plant, the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected fossil fuel power plants and by the addition of new generation sources.

Baseline emissions include only CO₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. The baseline emissions are to be calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

Where:

BE_y = Baseline emissions in year y (t CO₂/yr)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EF_{grid,CM,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of "TOOL07: Tool to calculate the emission factor for an electricity system" (tCO₂/ MWh).

The value taken is 0.397 tCO₂e/MWh¹⁶ sourced from the IFI Harmonized Emission Grid Factors dataset v.3.2 for the country of Malawi. In case of Greenfield power plant development, the calculation of the electricity system's combined margin (CM) shall be based on the following equation, according to the methodology:

$$EG_{PJ,y} = EG_{Facility,y}$$

4.2 Project Emissions

The project activity involves solar PV based renewable power generation which has no emissions over its lifetime,

hence $PE_y = 0$.

4.3 Leakage Emissions

As this current project is a greenfield Solar PV project, in line with ACM0002, no leakage is considered for this project.

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

The equations used for Net GHG Emission Reductions are calculated as follows:

$$EEE_{yy} = BEE_{yy} - LEE_{yy} - PEE_{yy}$$

Where

ER_y = Emission reductions in year y (tCO₂e)

BE_y = Baseline emissions in year y (tCO₂e)

PE_y = Project emissions in year y (tCO₂e)

LE_y = Leakage emissions in year y (tCO₂e)

Since $LE_y = 0$ and $PE_y = 0$

$ER_y = BE_y$

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCUs (tCO ₂ e)	Estimated removal VCUs (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
29-July-2023 to 31-December-2023	9,227	0	0	9,227	-	9,227
01-January-2024 to 31-December-2024	22,148	0	0	22,148	-	22,148
01-January-2025 to 31-December-2025	22,148	0	0	22,148	-	22,148
01-January-2026 to 31-December-2026	22,148	0	0	22,148	-	22,148
01-January-2027 to 31-December-2027	22,148	0	0	22,148	-	22,148
01-January-2028 to 31-December-2028	22,148	0	0	22,148	-	22,148
01-January-2029 to 31-December-2029	22,148	0	0	22,148	-	22,148
01-January-2030 to 29-July-2030	22,920	0	0	22,920	-	22,920
Total	155,036	0	0	155,036	-	155,036

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	EF _{grid,CM,y}
Data unit	tCO ₂ e/MWhr
Description	Emission factor for Malawian Grid taken from UN dataset
Source of data	UNFCCC IFI Harmonized Grid Emission Factors V3.2 Standardized Baseline
Value applied	0.397 tCO ₂ e/MWh
Justification of choice of data or description of measurement methods and procedures applied	Combined Margin emissions from IFI Harmonized Grid Emission Factors version 3.2 dataset is the only relevant and complete dataset available.
Purpose of data	Calculation of baseline emissions
Comments	Not applicable

5.2 Data and Parameters Monitored

Data / Parameter	EG _{PJ,y}
Data unit	MWh/yr
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y
Source of data	Direct measurement of electricity supplied to the grid using project meters installed
Description of measurement methods and procedures to be applied	The Net electricity supplied to the grid by the project activity will be calculated based on the energy delivered to & received from the grid monitored and measured by Main Meter and backup Meter. Meter Make: CeWe Pro meter 100 Meter Model: Advanced
Frequency of monitoring/recording	Continuous monitoring and monthly recording
Value applied	MWh
Monitoring equipment	CeWe Pro Meter 100 advanced of 0.2 class

QA/QC procedures to be applied	Calibration of all the meters will be undertaken at yearly intervals and faulty meters will be duly replaced immediately. Calibration Frequency: Once a year or as per the PPA. The data will be cross checked with sales receipts
Purpose of data	Calculation of project emissions
Calculation method	Visual confirmation from main meter and check meter
Comments	Not applicable

5.3 Monitoring Plan

The monitoring plan for this project is developed in accordance with methodological requirements, project activity instances will be renewable energy-based power generation projects. The project instance is a grid-connected solar power project being implemented in Nkhotakota, Malawi. The monitoring plan, which will be implemented by the project participant describes the monitoring organization, parameters to be monitored, monitoring practices, quality assurance, quality control procedures, data storage and archiving.

Roles and responsibilities of Facilities, General Manager: Overall functioning and maintenance of the project activity, collect and manage all data and its archiving and shall coordinate with the O&M operator and project proponent.

Project Personnel: hired and trained to maintain the project facilities. The project personnel may be drawn from the local population for maintenance and routine management.

QA/QC for Meters: Project meter will be maintained and checked by Project Proponent and check meter will be maintained and checked by the Malawian ESCOM. The meters are sealed and can only be opened in the presence of representatives from both the project proponent and ESCOM.

The project proponent shall check the reading from the main meter on the last day of the month to check net electrical output.

Meter type	Serial Number	Make and Model	Accuracy class	Date of installation
Main Meter	WP028188	CeWe Pro meter-Advance	0.2	19 th April, 2023
Check Meter	WP028189	CeWe Pro meter-Advance	0.2	19 th April, 2023

The records will be cross-checked with the records of sold electricity i.e., with the invoice. The meters will be calibrated at least once a year by the Project Proponent as per the terms of the PPA and TCA.

Data Measurement of Project: The export and import energy will be measured continuously using Main & check meter installed at inter connection point of project(as mention above). Net export will be calculated from the export & Import electricity readings of Main meter installed at project interconnection point and shall be taken on monthly basis by authorized officers of Project Proponent in the presence of representative of Malawian Grid and Project proponent if required.

Data collection: Export & Import readings from the meters will be collected under the supervision of the authorized representatives of PP. The net electricity supplied to the grid would be calculated based on Main Meter readings. Data would be recorded and stored in photographic and paper format not less than 7 years post end of PPA.

Emergency preparedness the project activity will not result in any unidentified activity that can result in substantial emissions from the project activity. No need for emergency preparedness in data monitoring is visualized. In the unlikely event of failure of all Main meter installed at site substation backup meter reading shall be used and all the faulty meters are required to be repaired or replaced simultaneously, the export readings from Main & backup Meters installed at the inter-connection point at the project site will be used for monitoring of net electricity exported to the grid.

Personnel training: In order to ensure the proper functioning of the project activity and accurate monitoring of emission reductions, the staff will be trained. The plant helpers will be trained in equipment operation, data recording, reports writing, operation and maintenance and emergency procedures in compliance with the monitoring plan.

APPENDIX 1: COMMERCIALY SENSITIVE INFORMATION

Not Applicable.

[illegible]

Image 2 Images of LSC invitations pasted in prominent public spaces



Image 3 Execution of LSC on 12th Oct 2023



Image 4 Grievance boxes installed at public spaces

APPENDIX 3: PROJECT MAIN METER AND CHECK METER



Image 5 Main Meter and Check Meter installed at project site